



Project #1: Desk Buddy

SWE Projects - Spring 2025

Project Lead: Tanya Koganti





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01

Resources/ Materials

Know all your components !

Resources

- Desk Buddy w/o Arms
<https://www.instructables.com/This-Little-Robots-Eyes-Reveal-Your-Rooms-Health/>
- Desk Buddy 2.0
<https://www.instructables.com/Gus-20-Uses-Emotions-to-Indicate-Room-Health/>
- Arms Inspo
<https://www.instructables.com/Cute-Desktop-Companion-Robot-TN24/>
- Additional Features
https://youtube.com/shorts/70Q-09GLxWw?si=f_ZWhWSxQvEXul8j

Online Competition ends May 12th 2025:

<https://www.instructables.com/contest/robots25/>

Customization IDEAS

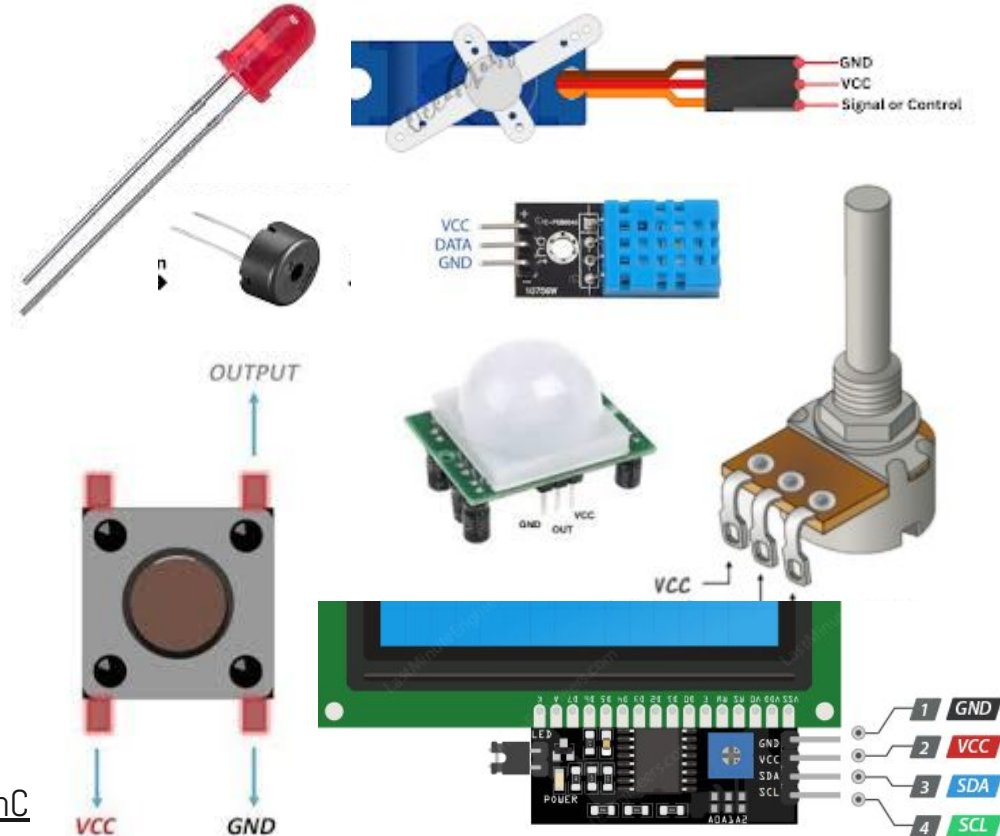
- Weather buddy
- Study Buddy
- Pomodoro Timer
- Pet Buddy
- Mood Matching Buddy
- Motion Security Buddy
- ANYTHING within materials (BOM)

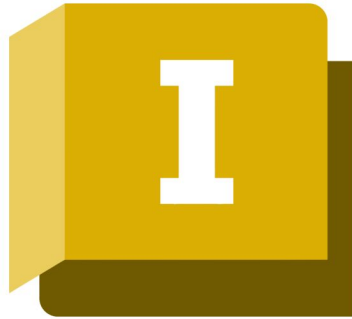
Know your Materials

- 3D Printing filament (3d Printer)
- Arduino Nano
- LCD Screen (16x2) w/ I2C Interface Module
- Speaker/Buzzer
- Power Supply (Battery Pack or USB Adapter)
- Resistors (4.7 Kohms)
- Breadboards & Wires
- Buttons
- Servo Motors (2)
- 1) Temperature/Humidity Sensor e.g.DHT11
- 2) PIR Motion Sensor e.g.HC-SR501

Bill of Materials:

<https://docs.google.com/spreadsheets/d/1qcdRV8rWDhCwpYHedvZe0XC4irzabry5pV66Vg0NF-k/edit?usp=sharing>





02

Design/CAD/ Mechanical

Plan out what you want to build !



Three Stages of Mechanical Prototyping



1. Design

Create sketches and plan the Desk Buddy's features, layout, and interactions.



2. CAD

Develop a 3D model and integrated all electronic and mechanical components.



3. 3D Print

Fabricate and test the prototype, refining the design through multiple iterations.

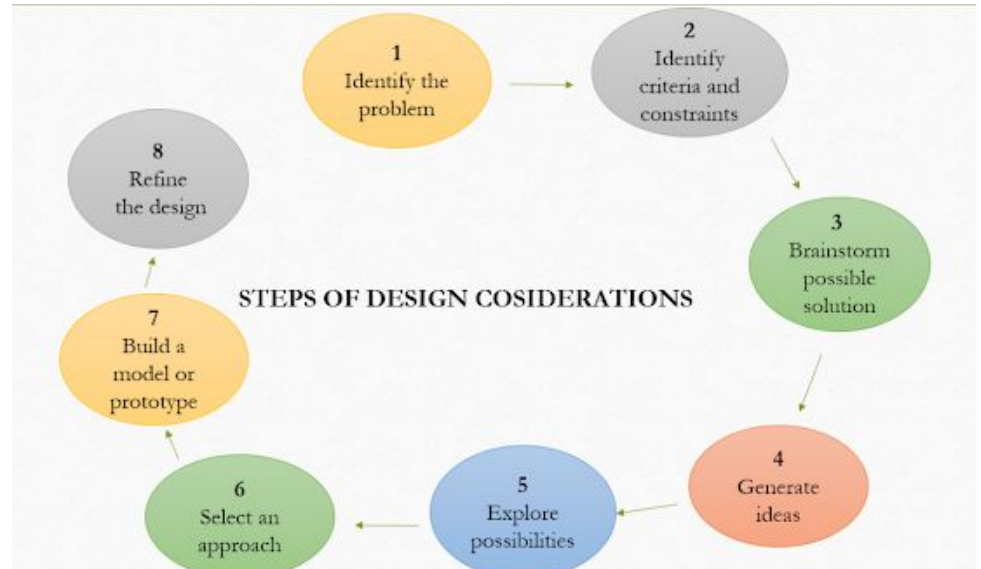
Engineering Design Constraints & Considerations

Design Considerations

Where will each component fit?
 How do we route wires cleanly?
 Will the design be easy to assemble?
 Can the structure support the hardware?
 How do we make the design compact and functional?

Mechanical Planning

Measuring motors, sensors, and batteries
 Planning mounting locations
 Designing around real component dimensions
 Balancing aesthetics and functionality



Students learned that engineering design starts long before fabrication.

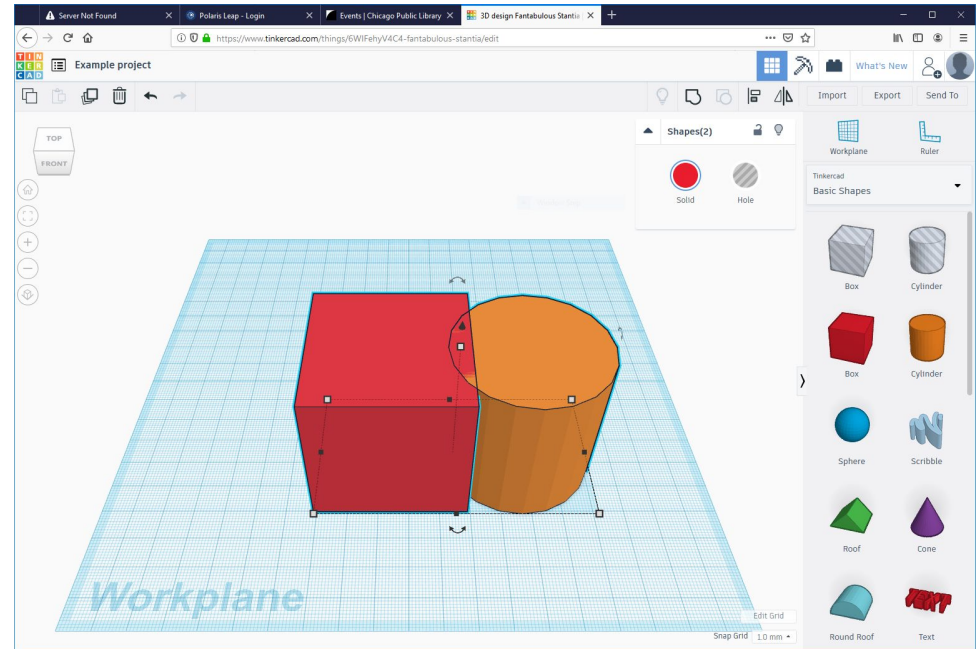
CAD - Finalize Design/3D Printing

CAD Development

- Creating initial sketches and part layouts
- Iterating designs based on component fit
- Learning tolerances and clearances
- Preparing models for fabrication

3D Printing & Prototyping Planning

- Exporting STL files for printing
- Testing printed parts for fit/function
- Adjusting designs after failed prints
- Understanding real-world manufacturing constraints
- Turning ideas into physical prototypes



Members gained hands-on experience with the full prototyping workflow.



Inventor or TinkerCAD

Inventor - AutoDesk



- Similar to SolidWorks
- Industry-standard CAD software
- Great introduction to mechanical design workflows
- Helps build technical portfolio projects
- Best for students wanting deeper CAD experience

Download:

<https://www.autodesk.com/education/edu-software/overview#INVPROSA>

TinkerCAD - AutoDesk



- Beginner-friendly CAD platform
- Easy entry point for first-time designers
- Quick learning curve for workshops/projects
- Great for Arduino + 3D printing integration
- Helps students gain confidence before advanced CAD

3D Design TinkerCAD:

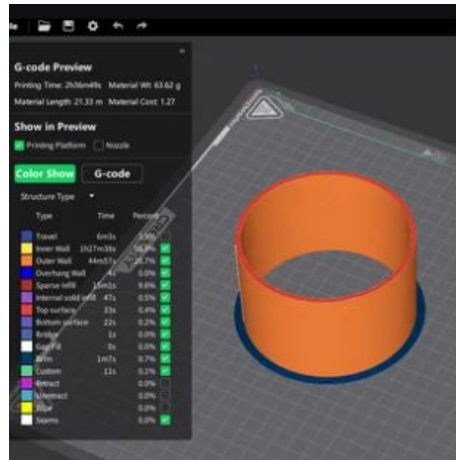
<https://www.tinkercad.com/3d-design>

These softwares chosen to make CAD more accessible for students of all experience levels.

3D Printing Slicer Software (CAD → 3D Printer)

CAD are in .STL Files
3D Printers can only read G-code

Slicer software converts .STL files to G-code instructions



Basic Slicer Workflow

- Import .STL file into slicer software
- Position model flat on the print bed
- Optimize orientation for stronger/more successful prints
- Scale or resize if necessary
- Add supports for overhangs or floating sections
- Check layer preview before printing
- Export final G-code to printer

Good Printing Practices

- Minimize unnecessary supports
- Avoid tall unstable orientations
- Keep large flat surfaces on the print bed
- Watch for unsupported overhangs
- Verify print time and material usage before printing



Ender3 Printer → Creality Slicer

Different printers may use different slicer software.

SWE projects used the Ender 3 printer with Creality Slicer.

Download Slicer Software "Creality Print":

<https://www.creality.com/download/ender-3-3d-printer>



03

Electrical/ Schematics

Wire it up, No wires = No programming...

Black Box Diagrams - Visualization

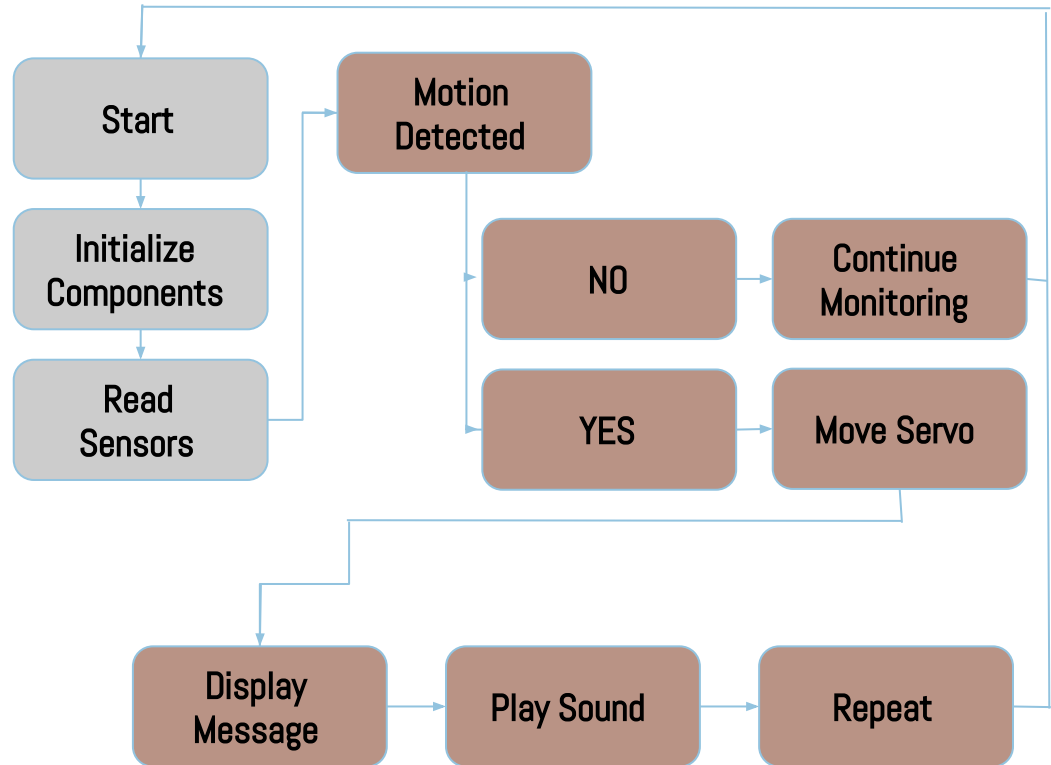
Black Box Diagram



Explains **what** the system does by showing inputs, outputs, and the system boundary.

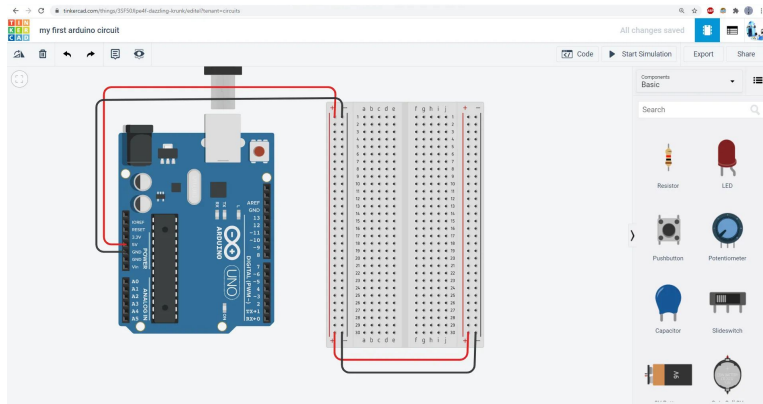
Flow Diagram/Flow Chart

Illustrates the sequence of operations and decision-making within the program. It helps visualize how the Desk Buddy responds to sensor inputs and controls its outputs.



Breadboard

TinkerCAD



Resources

Reading:

<https://www.build-electronic-circuits.com/breadboard/>

Videos:

https://youtu.be/xs8No_rg40U?si=nRS4d0bBDzABbNn

Intro Slides: https://docs.google.com/presentation/d/1yD5nXMJWPMtZPHhtXyCd_YX-FAF2YCUzgcZnmeqP8_c/edit?usp=sharing



The screenshot displays a PCB design tool interface with a schematic diagram of a custom board. Two microcontroller units, labeled "Bank2" and "Bank0", are shown at the top. Bank2 has pins RGB0 (39), RGB1 (40), and RGB2 (41). Bank0 has pins RGB2 (41) and RGB2 (41). Below them, the SPI interface is detailed with pin numbers: Bank2 SPI MOSI-I00 (14), Bank2 SPI MISO-I01 (17), Bank2 SPI SCK (16), Bank2 SPI CS (16), Bank0 SPI VCCIO1 (22), Bank0 SPI I0B_13b-LEDG (13), Bank0 SPI I0B_16a (10), Bank0 SPI I0B_18a (11), Bank0 SPI I0B_20a (12), Bank0 SPI I0B_22a (12), Bank0 SPI I0B_23b (12), Bank0 SPI I0B_24a (13), Bank0 SPI I0B_25b_G3 (13), Bank0 SPI I0B_25b_I0D3 (13), Bank0 SPI I0B_29b (10), Bank0 SPI I0B_31b (18), Bank0 SPI CDONE (7), Bank0 SPI CRESSET_B (8).

The schematic shows a power supply section on the left with a +V2 rail and a +V3 rail. A 100nF capacitor is connected between +V2 and GND. A 100k resistor is connected between +V3 and GND. Various other components like capacitors (C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, C26, C27, C28, C29, C30, C31, C32, C33, C34, C35, C36, C37, C38, C39, C40, C41, C42, C43, C44, C45, C46, C47, C48, C49, C50, C51, C52, C53, C54, C55, C56, C57, C58, C59, C60, C61, C62, C63, C64, C65, C66, C67, C68, C69, C70, C71, C72, C73, C74, C75, C76, C77, C78, C79, C80, C81, C82, C83, C84, C85, C86, C87, C88, C89, C90, C91, C92, C93, C94, C95, C96, C97, C98, C99, C100, C101, C102, C103, C104, C105, C106, C107, C108, C109, C110, C111, C112, C113, C114, C115, C116, C117, C118, C119, C120, C121, C122, C123, C124, C125, C126, C127, C128, C129, C130, C131, C132, C133, C134, C135, C136, C137, C138, C139, C140, C141, C142, C143, C144, C145, C146, C147, C148, C149, C150, C151, C152, C153, C154, C155, C156, C157, C158, C159, C160, C161, C162, C163, C164, C165, C166, C167, C168, C169, C170, C171, C172, C173, C174, C175, C176, C177, C178, C179, C180, C181, C182, C183, C184, C185, C186, C187, C188, C189, C190, C191, C192, C193, C194, C195, C196, C197, C198, C199, C200, C201, C202, C203, C204, C205, C206, C207, C208, C209, C210, C211, C212, C213, C214, C215, C216, C217, C218, C219, C220, C221, C222, C223, C224, C225, C226, C227, C228, C229, C230, C231, C232, C233, C234, C235, C236, C237, C238, C239, C240, C241, C242, C243, C244, C245, C246, C247, C248, C249, C250, C251, C252, C253, C254, C255, C256, C257, C258, C259, C260, C261, C262, C263, C264, C265, C266, C267, C268, C269, C270, C271, C272, C273, C274, C275, C276, C277, C278, C279, C280, C281, C282, C283, C284, C285, C286, C287, C288, C289, C290, C291, C292, C293, C294, C295, C296, C297, C298, C299, C300, C301, C302, C303, C304, C305, C306, C307, C308, C309, C310, C311, C312, C313, C314, C315, C316, C317, C318, C319, C320, C321, C322, C323, C324, C325, C326, C327, C328, C329, C330, C331, C332, C333, C334, C335, C336, C337, C338, C339, C340, C341, C342, C343, C344, C345, C346, C347, C348, C349, C350, C351, C352, C353, C354, C355, C356, C357, C358, C359, C360, C361, C362, C363, C364, C365, C366, C367, C368, C369, C370, C371, C372, C373, C374, C375, C376, C377, C378, C379, C380, C381, C382, C383, C384, C385, C386, C387, C388, C389, C390, C391, C392, C393, C394, C395, C396, C397, C398, C399, C400, C401, C402, C403, C404, C405, C406, C407, C408, C409, C410, C411, C412, C413, C414, C415, C416, C417, C418, C419, C420, C421, C422, C423, C424, C425, C426, C427, C428, C429, C430, C431, C432, C433, C434, C435, C436, C437, C438, C439, C440, C441, C442, C443, C444, C445, C446, C447, C448, C449, C450, C451, C452, C453, C454, C455, C456, C457, C458, C459, C460, C461, C462, C463, C464, C465, C466, C467, C468, C469, C470, C471, C472, C473, C474, C475, C476, C477, C478, C479, C480, C481, C482, C483, C484, C485, C486, C487, C488, C489, C490, C491, C492, C493, C494, C495, C496, C497, C498, C499, C500, C501, C502, C503, C504, C505, C506, C507, C508, C509, C510, C511, C512, C513, C514, C515, C516, C517, C518, C519, C520, C521, C522, C523, C524, C525, C526, C527, C528, C529, C530, C531, C532, C533, C534, C535, C536, C537, C538, C539, C540, C541, C542, C543, C544, C545, C546, C547, C548, C549, C550, C551, C552, C553, C554, C555, C556, C557, C558, C559, C560, C561, C562, C563, C564, C565, C566, C567, C568, C569, C570, C571, C572, C573, C574, C575, C576, C577, C578, C579, C580, C581, C582, C583, C584, C585, C586, C587, C588, C589, C590, C591, C592, C593, C594, C595, C596, C597, C598, C599, C600, C601, C602, C603, C604, C605, C606, C607, C608, C609, C610, C611, C612, C613, C614, C615, C616, C617, C618, C619, C620, C621, C622, C623, C624, C625, C626, C627, C628, C629, C630, C631, C632, C633, C634, C635, C636, C637, C638, C639, C640, C641, C642, C643, C644, C645, C646, C647, C648, C649, C650, C651, C652, C653, C654, C655, C656, C657, C658, C659, C660, C661, C662, C663, C664, C665, C666, C667, C668, C669, C670, C671, C672, C673, C674, C675, C676, C677, C678, C679, C680, C681, C682, C683, C684, C685, C686, C687, C688, C689, C690, C691, C692, C693, C694, C695, C696, C697, C698, C699, C700, C701, C702, C703, C704, C705, C706, C707, C708, C709, C710, C711, C712, C713, C714, C715, C716, C717, C718, C719, C720, C721, C722, C723, C724, C725, C726, C727, C728, C729, C730, C731, C732, C733, C734, C735, C736, C737, C738, C739, C740, C741, C742, C743, C744, C745, C746, C747, C748, C749, C750, C751, C752, C753, C754, C755, C756, C757, C758, C

[illegible]

Sensors



PIR Sensor (Motion) Passive Infrared Sensor

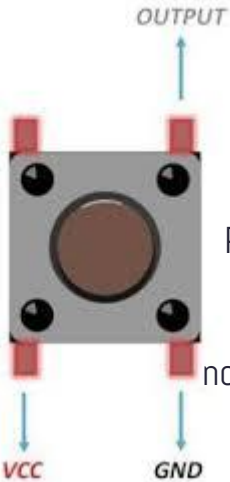
Detects movement near the Desk Buddy. When motion is sensed, the Desk Buddy "wakes up" and begins interacting with the user through sounds, messages, or movement.



Temperature Sensor

Measures the surrounding temperature and allows the Desk Buddy to display environmental information or react differently based on room conditions.

Components (Input)

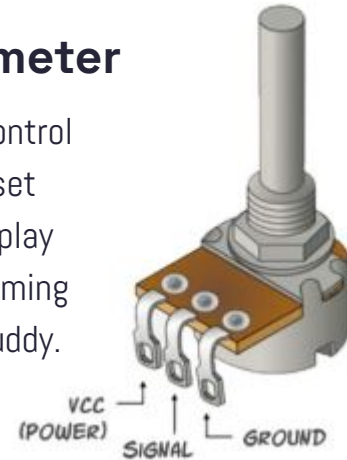


Push Button

Provides direct user input to the Desk Buddy. It can be used to change modes, dismiss notifications, trigger animations, or interact with menu options.

Potentiometer

Acts as an adjustable control knob. It can be used to set preferences such as display brightness, volume, or timing settings for the Desk Buddy.



Components (Output)

Passive Buzzer



Buzzer (Passive)

Produces customizable tones and melodies using different frequencies. It allows the Desk Buddy to play sounds for reminders, alerts, and user interactions.

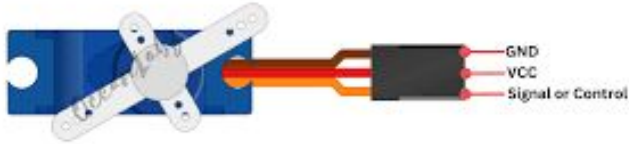
Active Buzzer



Buzzer (Active)

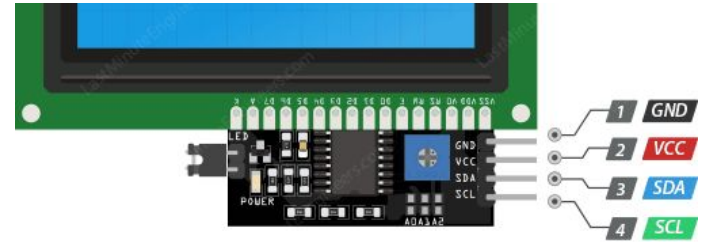
Generates a simple alert sound when powered. It is used for quick notifications, alarms, and status indicators.

Components



Servo Motor

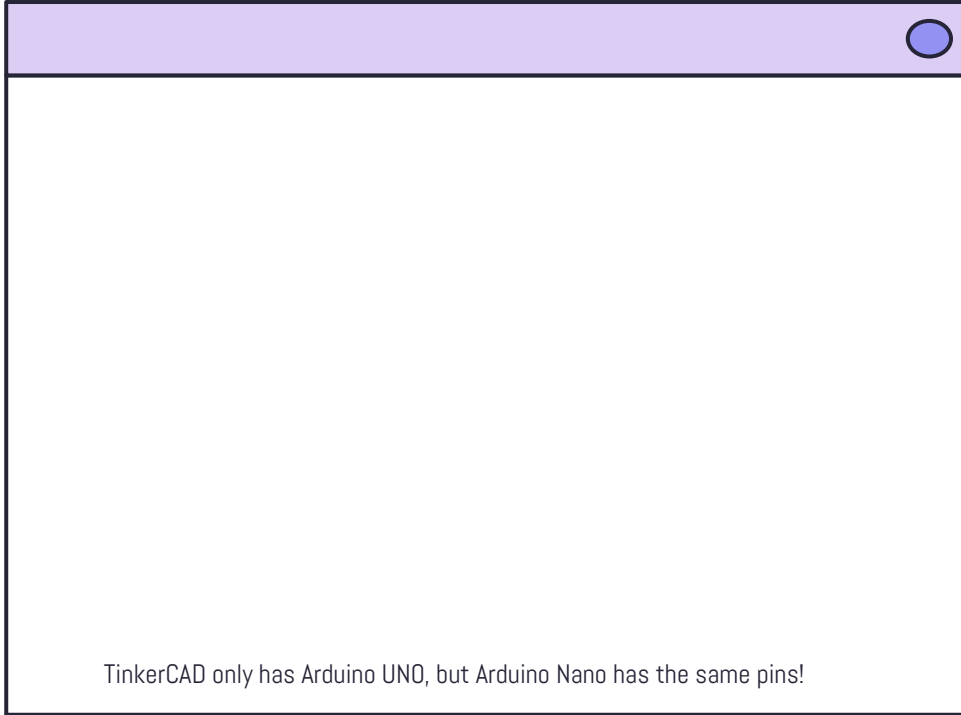
Controls physical movement of the Desk Buddy. It can rotate arms, tilt the head, or create expressive motions that make the project feel more interactive and lifelike.



LCD Screen (w/ I2C)

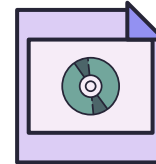
Provides visual feedback by displaying messages, temperature readings, reminders, system status, or animated expressions for the Desk Buddy.

Sample Circuit - Customize your Own !!



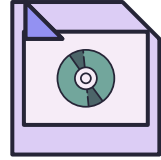
Nano

main microcontroller,
reading sensor inputs
and controlling outputs



Jumper Wires

Connect the Arduino,
breadboard, and
electronic components



Power

Distributes 5V and
GND from the Arduino
to the breadboard,



Resistors

Limit current to
protect components
and ensure safety



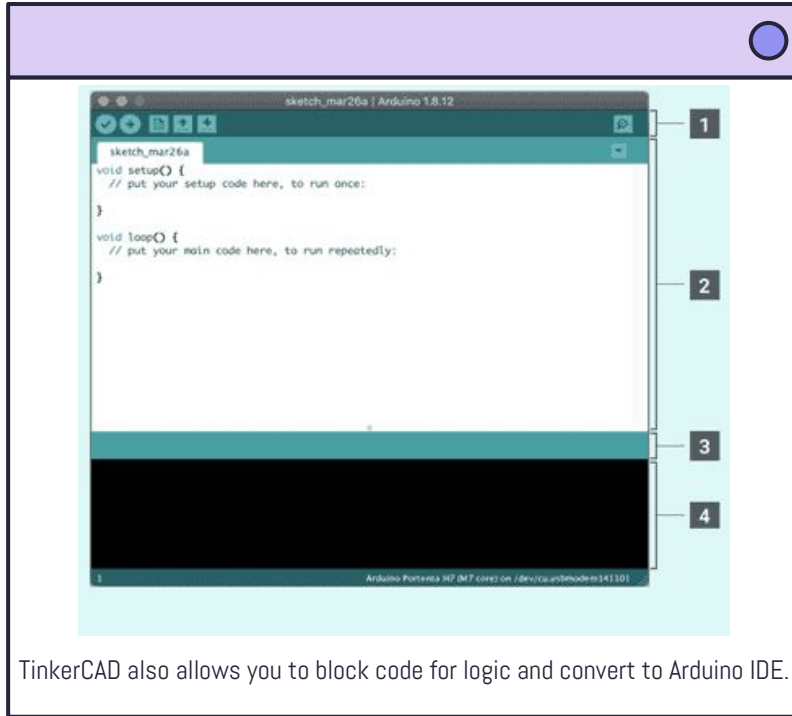
04

Programming

Watch your build come to life !!

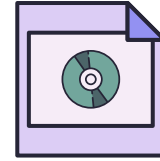


Arduino IDE (C++) -



Download

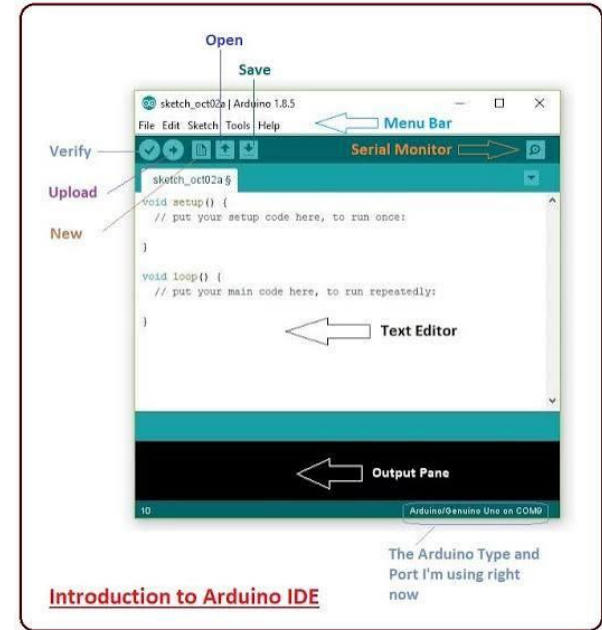
<https://support.arduino.cc/hc/en-us/articles/360019833020-Download-and-install-Arduino-IDE>



Arduino IDE Function Library

<https://docs.arduino.cc/language-reference/>

Arduino IDE - Must Know



Arduino IDE Intro: <https://docs.arduino.cc/software/ide-v2/tutorials/getting-started-ide-v2/>



Arduino IDE - Parts

The screenshot shows the Arduino IDE window titled "Blink | Arduino 1.0". The code editor contains the following text:

```
/*
 * Blink
 * Turns on an LED on for one second, then off for one second, repeatedly.
 *
 * This example code is in the public domain.
 */

void setup() {
  // initialize the digital pin as an output.
  // Pin 13 has an LED connected on most Arduino boards:
  pinMode(13, OUTPUT);
}

void loop() {
  digitalWrite(13, HIGH); // set the LED on
  delay(1000);            // wait for a second
  digitalWrite(13, LOW);  // set the LED off
  delay(1000);            // wait for a second
}
```

At the bottom of the window, a status bar indicates "1" and "Arduino Uno on /dev/tty.usbmodem621".

void setup ()

Setup Function

Runs once when the Arduino powers on. Used to initialize components such as sensors, servos, buzzers, and the LCD display before the Desk Buddy begins operating.

void loop ()

Main Program Loop

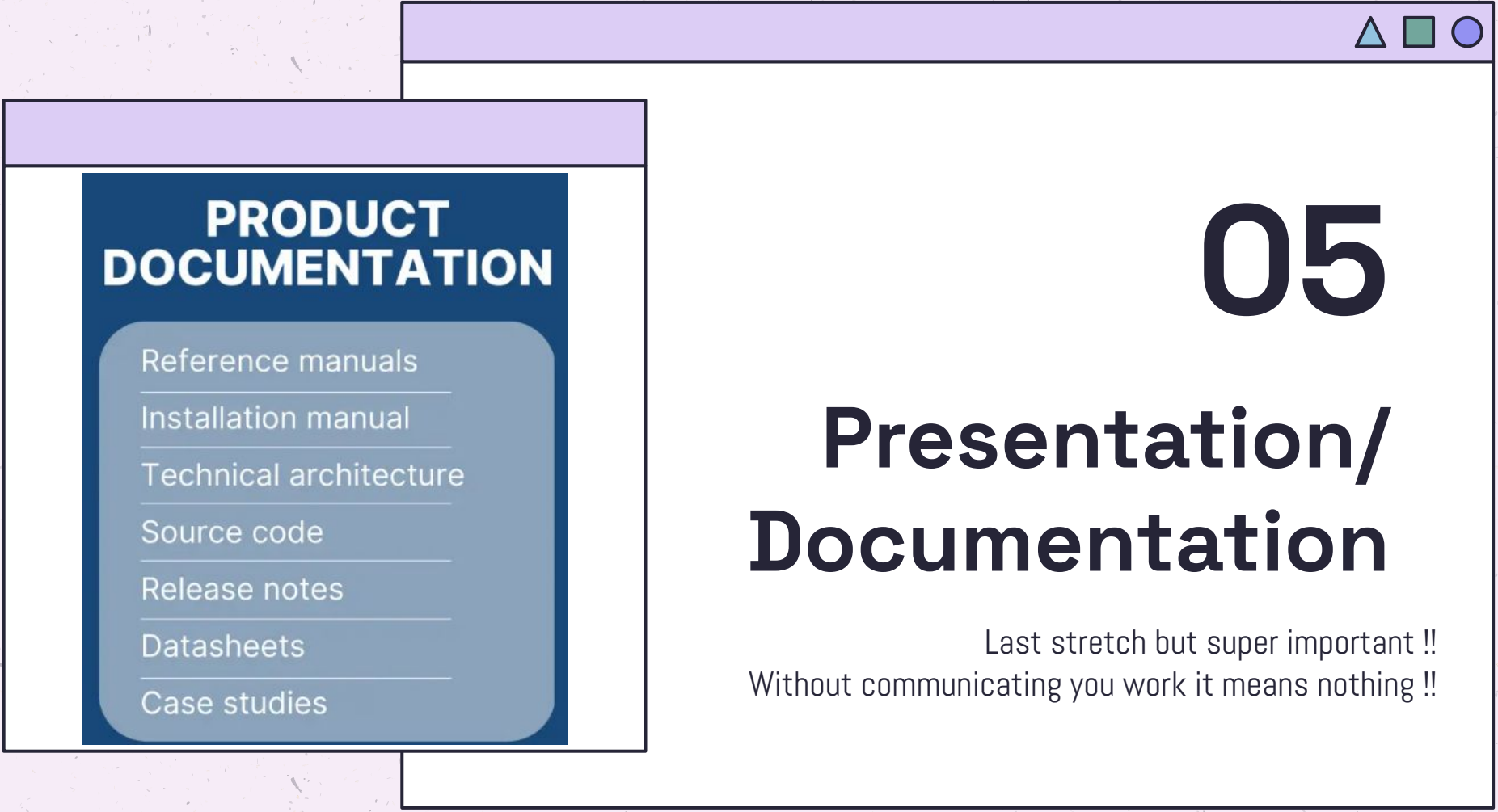
Runs continuously after `setup()`. It repeatedly checks sensor inputs and updates the Desk Buddy's actions, displays, sounds, and movements.

Variables

Store information used throughout the program, such as temperature readings, motion detection status, servo positions, and user settings.

Functions

Reusable blocks of code that perform specific tasks. Functions help organize the program and simplify actions like displaying messages, playing sounds, or moving servos.



PRODUCT DOCUMENTATION

Reference manuals

Installation manual

Technical architecture

Source code

Release notes

Datasheets

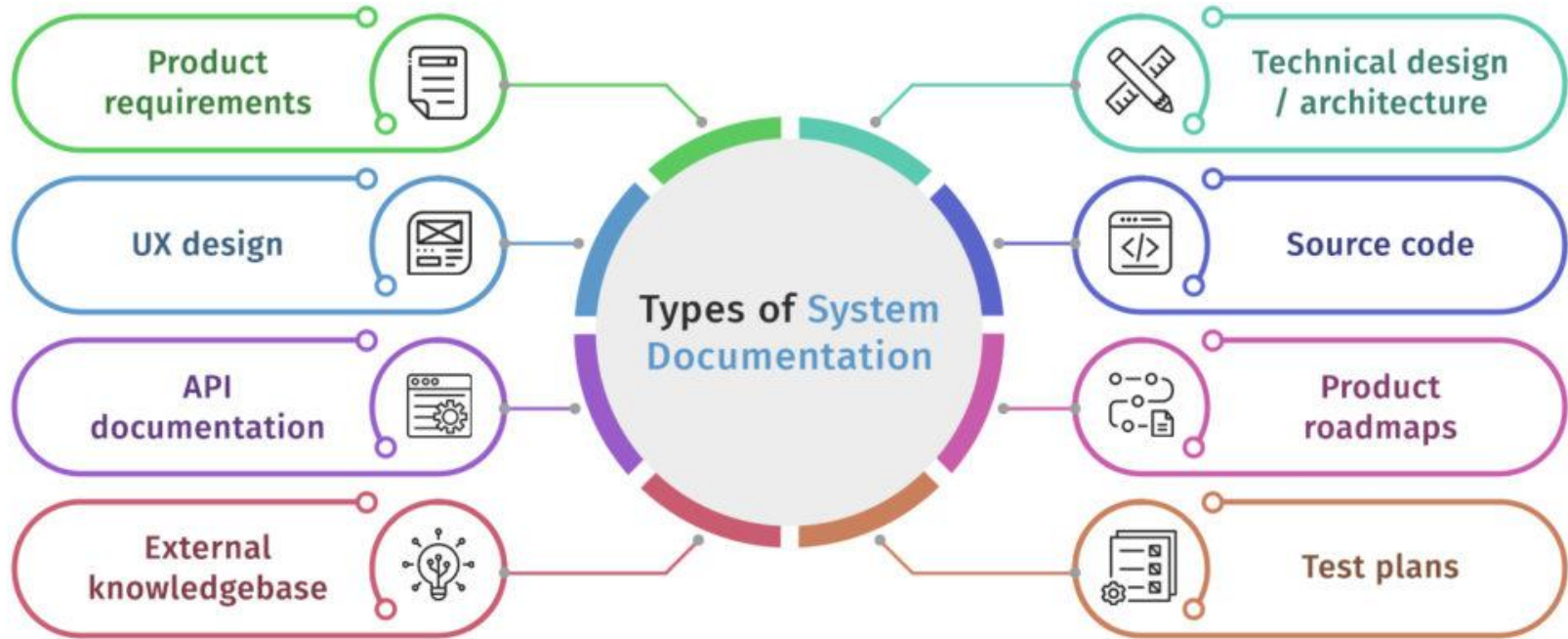
Case studies

05

Presentation/ Documentation

Last stretch but super important !!
Without communicating your work it means nothing !!

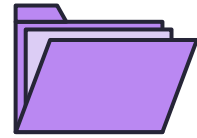
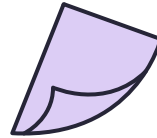
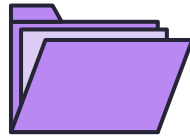
Product Based Documentation !!



<https://technicalwriterhq.com/documentation/product-documentation/>



Presentation - 4 parts



Design/CAD

CAD Files

Schematics

Circuit Schematic
File

Program w/ Comments

Arduino IDE Code
w/ Comments

Pictures of final Product

Pictures/Videos

Presentation

Presentation Template:

SWE Project Showcase (Demo) - Awards !!



Please have your
Project + Slides ready for our
Inaugural SWE Project Showcase !!

Thursday, April 24th
6:00 - 8:00pm
Arch 310

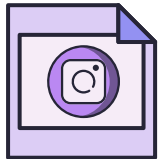


Thanks!

Do you have any questions?



Please Contact Me !!
Project.swe.uh@gmail.com
@Tako_124 on Discord



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All Projects accessible after the end of semester:

Feedback Form !!

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Troubleshooting Help - Common



LCD not Displaying

Check SDA/SCL
connections

Verify I2C address



Servo not work

Verify power connection

Check signal pin



PIR always active

Wait 30–60 seconds for
calibration

Further Learning

- D

Keep Learning !!

Future Improvements:

End each project with ideas like:

Future Improvements

- Wi-Fi connectivity
- Mobile app integration
- Voice assistant
- Battery-powered operation
- Additional sensors
- PCB instead of breadboard

This shows engineering doesn't stop at Ver 1.

